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Acoustic wave device, filter, multiplexer, and manufacturing method of acoustic wave device

Abstract

An acoustic wave device includes a substrate, lower and upper electrodes provided over the substrate, a piezoelectric film that is provided over the substrate, is interposed between the lower and upper electrodes, and has a pair of through holes that sandwich a resonance region therebetween in a first direction, are provided along the resonance region, and are connected to an air gap that is formed between the substrate and the lower electrode and overlaps the resonance region in the plan view, the lower and upper electrodes overlapping across the piezoelectric film in the resonance region, and additional films that are not provided in a central region of the resonance region in the plan view and are provided in respective edge regions, which are located on respective sides of the central region in a second direction substantially orthogonal to the first direction in the plan view, of the resonance region.

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2020-88680	12/2019	JP	N/A
2020-202465	12/2019	JP	N/A
2020-205533	12/2019	JP	N/A

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Ting Wu et al., “Application of Free Side Edges to Thickness Shear Bulk Acoust ic Resona tor on Lit hium Niobate for Suppression of Transverse Resonances”, Materials of the Se cond Res earch Meeting of a coustic Wave Element Technology Consortium, Mar. 8, 2021 (Mentioned in paragraph Nos. 3 and 5 of the application.). cited by applicant

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Background/Summary

FIELD

(1) A certain aspect of the present disclosure relates to an acoustic wave device, a filter, a multiplexer, and a manufacturing method of an acoustic wave device.

BACKGROUND

(2) As filters and duplexers for high frequency circuits of wireless terminals such as mobile phones, filters and duplexers using piezoelectric thin film resonators are known. The piezoelectric thin film resonator includes a piezoelectric film, and a lower electrode and an upper electrode sandwiching the piezoelectric film therebetween. The region where the lower electrode and the upper electrode are opposite to each other across the piezoelectric film is a resonance region where the acoustic wave resonates.

(3) In the piezoelectric thin film resonator, when an acoustic wave is reflected at the periphery of the resonance region and a standing wave is formed in the resonance region, it becomes an unnecessary spurious emission. Therefore, it is known to reduce spurious emissions by forming an additional film in an edge region within the resonance region as disclosed in for example, Japanese Patent Application Laid-Open Nos. 2008-42871, 2020-88680, and 2020-202465 (Patent Documents 1 to 3), and to reduce spurious emissions by providing an air gap on the side of the resonance region as disclosed in, for example, Ting Wu et al., “Application of Free Side Edges to Thickness Shear Bulk Acoustic Resonator On Lithium Niobate for Suppression of Transverse Resonance”, Materials of the Second Research Meeting of Acoustic Wave Element Technology Consortium, Mar. 8, 2021 (Non-patent document 1). It is also known that spurious emissions are reduced by providing irregularities on a pair of surfaces reflecting the acoustic wave of a multilayered film including the lower electrode, the piezoelectric film, and the upper electrode as disclosed in, for example, Japanese Patent Application Laid-Open No. 2020-202465 (Patent Document 4). Further, it is known to use a lithium niobate film or a lithium tantalate film as the piezoelectric film (for example, Patent Documents 1 to 3 and Non-Patent Document 1).

RELATED ART DOCUMENTS

Patent Documents

(4) Japanese Patent Application Laid-Open No. 2008-42871 Japanese Patent Application Laid-Open No. 2020-88680 Japanese Patent Application Laid-Open No. 2020-205533 Japanese Patent Application Laid-Open No. 2020-202465

Non-Patent Documents

(5) Ting Wu et al., “Application of Free Side Edges to Thickness Shear Bulk Acoustic Resonator On Lithium Niobate for Suppression of Transverse Resonances”, Materials of the Second Research Meeting of Acoustic Wave Element Technology Consortium, Mar. 8, 2021

SUMMARY

(6) As described in Patent Documents 1 to 3, spurious emissions can be reduced by forming an additional film in the edge region within the resonance region. However, there is still room for improvement in terms of reducing spurious emissions while reducing the size of the acoustic wave device.

(7) It is an object of the present disclosure to reduce the size of an acoustic wave device and reduce spurious emissions.

(8) The present disclosure provides, in one aspect, an acoustic wave device including: a substrate; a lower electrode and an upper electrode provided over the substrate; a piezoelectric film that is provided over the substrate, at least a part of the piezoelectric film being interposed between the lower electrode and the upper electrode thereby defining a resonance region in a plan view where the lower electrode and the upper electrode overlap with each other across said at least a part of the piezoelectric film, the piezoelectric film having a pair of through holes, the pair of through holes sandwiching the resonance region therebetween in a first direction in the plan view, being provided along the resonance region, and being connected to an air gap, the air gap being formed between

the substrate and the lower electrode and overlapping with the resonance region in the plan view; and additional films that are not provided in a central region of the resonance region in the plan view and are provided in respective edge regions, which are located on respective sides of the central region in a second direction substantially orthogonal to the first direction in the plan view, of the resonance region.

(9) In the above acoustic wave device, the lower electrode and the upper electrode may excite thickness-shear vibration in the piezoelectric film, and the second direction may be substantially parallel to a vibration direction of the thickness-shear vibration.

(10) In the above acoustic wave device, a substantially entire circumference of the resonance region may be surrounded by the additional films and the pair of through holes in the plan view.

(11) In the above acoustic wave device, in the plan view, the pair of through holes may be in contact with the resonance region at respective sides of the resonance region in the first direction.

(12) In the above acoustic wave device, the resonance region may have a substantially rectangular shape in the plan view, and sides opposite to each other in the first direction of the resonance region may be defined by the pair of through holes.

(13) In the above acoustic wave device, the additional films may be longer than the resonance region in the first direction, and at least a part of each of the pair of through holes may be sandwiched between the additional films in the second direction in the plan view.

(14) In the above acoustic wave device, the additional films may be provided in a ring shape on at least one of the following surfaces: a first surface of the piezoelectric film on which the lower electrode is provided and a second surface of the piezoelectric film on which the upper electrode is provided, and the pair of through holes may be provided inside the ring-shaped additional films.

(15) The present disclosure provides, in another aspect, a filter including the above acoustic wave device.

(16) The present disclosure provides, in another aspect, a multiplexer including the above filter.

(17) The present disclosure provides, in another aspect, a manufacturing method of an acoustic wave device, the manufacturing method including: forming a lower electrode on a first surface of a piezoelectric film, the piezoelectric film further having a second surface opposite to the first surface; forming a first additional film on at least a tip portion of the lower electrode, the first additional film having a substantially rectangular shape in a plan view; forming a sacrificial layer covering the lower electrode and the first additional film over the first surface of the piezoelectric film; thereafter, bonding a resulting multilayer structure to a substrate such that the first surface of the piezoelectric film faces the substrate through the sacrificial layer and the first additional film; forming an upper electrode on the second surface of the piezoelectric film; forming a second additional film on at least a tip portion of the upper electrode, the second additional film having a substantially rectangular shape in a plan view; forming, in the piezoelectric film, a pair of through holes that sandwich a resonance region therebetween in a first direction and that are provided along the resonance region in a plan view, the resonance region being defined as a region where the lower electrode and the upper electrode overlap with each other across the piezoelectric film in the plan view, the first direction being a width direction of the lower electrode and the upper electrode; and removing the sacrificial layer by introducing an etching medium into the pair of through holes.

(18) In the above manufacturing method, the forming of the pair of through holes may include forming the pair of through holes so that a length of at least one of the pair of through holes in a second direction substantially orthogonal to the first direction is longer than a length of the resonance region in the second direction.

(19) In the above manufacturing method, the forming of the first additional film may include forming the first additional film that is provided from a tip portion of the lower electrode to a front side of the lower electrode, and has a side surface inclined with respect to a normal line of the first surface of the piezoelectric film in a cross-sectional view.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a plan view of a piezoelectric thin film resonator in accordance with a first embodiment;

(2) FIG. **2**A is a cross-sectional view taken along line A-A in FIG. **1**, and FIG. **2**B is a cross-sectional view taken along line B-B in FIG. **1**;

(3) FIG. **3**A to FIG. **3**F are cross-sectional views (part **1**) illustrating a method of manufacturing the piezoelectric thin film resonator in accordance with the first embodiment;

(4) FIG. **4**A to FIG. **4**F are cross-sectional views (part **2**) illustrating the method of manufacturing the piezoelectric thin film resonator in accordance with the first embodiment;

(5) FIG. **5**A to FIG. **5**D are cross-sectional views (part **3**) illustrating the method of manufacturing the piezoelectric thin film resonator in accordance with the first embodiment;

(6) FIG. **6**A to FIG. **6**D are cross-sectional views (part **4**) illustrating the method of manufacturing the piezoelectric thin film resonator in accordance with the first embodiment;

(7) FIG. **7**A and FIG. **7**B illustrate crystal orientations in the case that the piezoelectric film is a lithium niobate film or a lithium tantalate film;

(8) FIG. **8** is a plan view of a piezoelectric thin film resonator in accordance with a comparative example;

(9) FIG. **9**A is a cross-sectional view taken along line A-A in FIG. **8**, and FIG. **9**B is a cross-sectional view taken along line B-B in FIG. **8**;

(10) FIG. **10**A to FIG. **10**C are plan views illustrating other examples of an air gap;

(11) FIG. **11** is a plan view of a piezoelectric thin film resonator in accordance with a second embodiment;

(12) FIG. **12**A is a cross-sectional view taken along line A-A in FIG. **11**, and FIG. **12**B is a cross-sectional view taken along line B-B in FIG. **11**;

(13) FIG. **13** is a plan view of a piezoelectric thin film resonator in accordance with a third embodiment;

(14) FIG. **14**A and FIG. **14**B are cross-sectional views of a piezoelectric thin film resonator in accordance with a fourth embodiment;

(15) FIG. **15** is a plan view of a piezoelectric thin film resonator in accordance with a fifth embodiment;

(16) FIG. **16**A is a cross-sectional view taken along line A-A in FIG. **15**, and FIG. **16**B is a cross-sectional view taken along line B-B in FIG. **15**;

(17) FIG. **17** is a circuit diagram of a filter in accordance with a sixth embodiment; and

(18) FIG. **18** is a block diagram of a duplexer in accordance with a seventh embodiment.

DETAILED DESCRIPTION

(19) Hereinafter, embodiments of the present invention will be described with reference to the accompanying drawings. In the embodiments, a piezoelectric thin film resonator will be described as an example of the acoustic wave device.

First Embodiment

(20) FIG. **1** is a plan view of a piezoelectric thin film resonator **100** in accordance with a first embodiment. FIG. **2**A is a cross-sectional view taken along line A-A in FIG. **1**, and FIG. **2**B is a cross-sectional view taken along line B-B in FIG. **1**.

(21) In FIG. **1**, for clarity of the drawing, a passivation film **28** is not illustrated, a resonance region **50** is hatched, and through holes **22** are illustrated by lines thicker than the others (the same applies to similar drawings hereinafter). The normal direction of a piezoelectric film **14** is defined as a Z direction, a vibration direction **60** of the thickness-shear vibration is defined as a Y direction, and the direction that is the plane direction of the piezoelectric film **14** and is perpendicular to the Y

direction is defined as an X direction. The X direction, the Y direction, and the Z direction do not necessarily correspond to the X axis, the Y axis, and the Z axis of the crystal orientation of the piezoelectric film **14**. When the crystal orientations are referred to, they are described as the X-axis orientation, the Y-axis orientation, and the Z-axis orientation, and are distinguished from the X direction, the Y direction, and the Z direction.

(22) As illustrated in FIG. **1**, FIG. **2**A, and FIG. **2**B, the piezoelectric thin film resonator **100** includes a substrate **10**, a lower electrode **12**, the piezoelectric film **14**, an upper electrode **16**, additional films **18***a* and **18***b*, terminal electrodes **24***a* and **24***b*, an insulating film **26**, and the passivation film **28**.

(23) The insulating film **26** is provided on the substrate **10** and has a recess **32**. The piezoelectric film **14** is provided on the insulating film **26** so as to cover the recess **32**. The upper and lower surfaces of the piezoelectric film **14** are substantially flat surfaces. The lower electrode **12** is provided on the lower surface of the piezoelectric film **14** so as to be positioned in the recess **32**. An air gap **30** is formed between the lower electrode **12** and the insulating film **26**. The upper electrode **16** is provided on the upper surface of the piezoelectric film **14**. A region where the lower electrode **12** and the upper electrode **16** overlap each other in a plan view with at least a part of the piezoelectric film **14** interposed therebetween is a resonance region **50**. The resonance region **50** is located so as to overlap the air gap **30** in a plan view. The air gap **30** is larger than the resonance region **50** in a plan view. In a plan view, the side surface of the lower electrode **12** and the side surface of the upper electrode **16** substantially coincide with each other, for example. The term “substantially coincide with” means that a difference is acceptable to the extent of manufacturing errors. The thickness of the lower electrode **12** and the thickness of the upper electrode **16** are, for example, about 20 nm to 150 nm. The thickness of the piezoelectric film **14** is, for example, about 200 nm to 1000 nm.

(24) The terminal electrode **24***a* is electrically connected to the lower electrode **12**, and the terminal electrode **24***b* is electrically connected to the upper electrode **16**. When high-frequency power is applied between the terminal electrode **24***a* and the terminal electrode **24***b* (i.e., between the lower electrode **12** and the upper electrode **16**), an acoustic wave having a displacement in a direction substantially orthogonal to the Z direction (i.e., the strain direction with respect to the thickness) is excited in the piezoelectric film **14** within the resonance region **50**. This vibration is referred to as thickness-shear vibration. The direction in which the displacement of the thickness-shear vibration is largest (the displacement direction of the thickness-shear vibration) is defined as the vibration direction **60** of the thickness-shear vibration. Here, the vibration direction **60** of the thickness-shear vibration is the Y direction. The wavelength of the acoustic wave is approximately twice the thickness of the piezoelectric film **14**. The planar shape of the resonance region **50** is substantially rectangular. The substantially rectangular shape has four substantially straight sides. As is clear from the above, the term “substantially rectangular” is not limited to a case in which the four sides are completely straight lines, and a case in which the shape is deformed from a rectangle to the extent that does not affect the device characteristics is also acceptable. Among the four sides, a pair of sides extend substantially in the Y direction (i.e., the vibration direction **60** of the thickness-shear vibration), and another pair of sides extend substantially in the X direction (i.e., the direction orthogonal to the vibration direction **60** of the thickness-shear vibration).

(25) The resonance region **50** has a central region **52** and edge regions **54***a* and **54***b* located on respective sides of the central region **52** in the Y direction. The edge regions **54***a* and **54***b* extend substantially in the X direction. The widths Wa and Wb of the edge regions **54***a* and **54***b* in the Y direction are substantially constant in the X direction.

(26) The additional film **18***a* is provided at least under the tip portion, which is located in the edge region **54***a*, of the lower electrode **12**, and is located in the recess **32** formed in the insulating film **26**. The additional film **18***a* has a substantially rectangular shape in a plan view. The substantially rectangular shape is not limited to a case in which each side is completely straight, and a case in

which the shape is deformed from a rectangle to the extent that does not affect the device characteristics is also acceptable. The additional film **18***a* is provided, for example, from the edge region **54***a* to a region **56***a* outside the resonance region **50**, and covers the edge surface of the lower electrode **12**. The thickness of the additional film **18***a* on the tip portion of the lower electrode **12** is, for example, about 10 nm to 70 nm. The additional film **18***b* is provided at least on the tip portion, which is located in the edge region **54***b*, of the upper electrode **16**. The additional film **18***b* has a substantially rectangular shape in a plan view. The substantially rectangular shape is not limited to a case in which each side is completely straight, and a case in which the shape is deformed from a rectangle to the extent that does not affect the device characteristics is also acceptable. The additional film **18***b* is provided, for example, from the edge region **54***b* to a region **56***b* outside the resonance region **50**, and covers the edge surface of the upper electrode **16**. The thickness of the additional film **18***b* on the tip portion of the upper electrode **16** is, for example, about 10 nm to 70 nm. As described above, the additional film **18***a* and the additional film **18***b* are provided apart from each other in the Y direction. The additional films **18***a* and **18***b* extend further outward than the resonance region **50**, for example, in the X direction.

(27) In the piezoelectric film **14**, a pair of the through holes **22** that sandwich the resonance region **50** therebetween in the X direction are formed, and are located along the resonance region **50** in the Y direction. In a plan view, the through holes **22** are in contact with the resonance region **50** on respective sides of the resonance region **50** in the X direction. That is, in a plan view, a pair of the through holes **22** define the sides opposite to each other in the X direction of the substantially rectangular resonance region **50**. Each through hole **22** has a substantially rectangular shape in a plan view, for example. The substantially rectangular shape is not limited to a case in which each side is completely straight, and a case in which the shape is deformed from a rectangle to the extent that does not affect the device characteristics is also acceptable Each through hole **22** is connected to the air gap **30**, and is communicated with the air gap **30**. Since the additional films **18***a* and **18***b* extend further outward than the resonance region **50** in the X direction, at least a part of each of the through holes **22** is sandwiched between the additional films **18***a* and **18***b* in the Y direction in a plan view. The additional films **18***a* and **18***b* and a pair of the through holes **22** are provided so as to surround the resonance region **50** in a plan view. The widths of the additional films **18***a* and **18***b* in the Y direction are, for example, about 1000 μm to 1500 μm and the widths of the through holes **22** in the X direction are, for example, about 1000 μm to 1500 μm. The length L of the air gap **30** between the side surface of the recess **32** of the insulating film **26** and the additional film **18***a* is preferably equal to or greater than 2λ (λ is the wavelength of the acoustic wave) to reduce the characteristic degradation.

(28) The passivation film **28** is provided so as to cover the upper surface of the piezoelectric film **14**, the surface of the upper electrode **16**, and the surface of the additional film **18***b*. The terminal electrode **24***a* penetrates through the passivation film **28** and the piezoelectric film **14** and is coupled to the lower electrode **12**. The terminal electrode **24***b* penetrates through the passivation film **28** and is coupled to the upper electrode **16**.

(29) The substrate **10** is, for example, a silicon substrate, a sapphire substrate, an alumina substrate, a spinel substrate, a quartz substrate, a crystal substrate, a glass substrate, a ceramic substrate, or a GaAs substrate. The insulating film **26** is, for example, a silicon oxide film, a silicon nitride film, or an aluminum oxide film. The piezoelectric film **14** is, for example, a monocrystalline lithium niobate film, a monocrystalline lithium tantalate film, or a crystal, and is, for example, a rotated Y-cut lithium niobate film or a rotated Y-cut lithium tantalate film. The lower electrode **12** and the upper electrode **16** are, for example, single-layer films of ruthenium (Ru), chromium (Cr), aluminum (Al), titanium (Ti), copper (Cu), molybdenum (Mo), tungsten (W), tantalum (Ta), platinum (Pt), rhodium (Rh), or iridium (Ir), or multilayered films of any combination thereof.

(30) The additional films **18***a* and **18***b* are, for example, metal films exemplified in the lower electrode **12** and the upper electrode **16** or insulating films such as silicon oxide films, silicon

nitride films, or aluminum oxide films. The additional film **18***a* and the additional film **18***b* may be formed of the same material or different materials. The passivation film **28** is an insulating film such as a silicon oxide film, a silicon nitride film, or an aluminum oxide film. The terminal electrodes **24***a* and **24***b* are metal layers including a gold layer, a copper layer or the like.
(31) Manufacturing Method
(32) FIG. **3**A to FIG. **6**D are cross-sectional views illustrating a method of manufacturing the piezoelectric thin film resonator **100** in accordance with the first embodiment. FIG. **3**A to FIG. **3**C, FIG. **4**A to FIG. **4**C, FIG. **5**A, FIG. **5**B, FIG. **6**A, and FIG. **6**B are cross-sectional views of the portion along line A-A in FIG. **1**. FIG. **3**D to FIG. **3**F, FIG. **4**D to FIG. **4**F, FIG. **5**C, FIG. **5**D, FIG. **6**C, and FIG. **6**D are cross-sectional views of the portion along line B-B in FIG. **1**.
(33) As illustrated in FIG. **3**A and FIG. **3**D, the lower electrode **12** is formed on a surface **15***a* (a first surface) of the piezoelectric film **14**. The piezoelectric film **14** here is a film (substrate) thicker than the piezoelectric film **14** illustrated in FIG. **2**A and FIG. **2**B. The lower electrode **12** is formed by depositing a film by, for example, sputtering, vacuum evaporation, or chemical vapor deposition (CVD), and then patterning the film into a desired shape by photolithography and etching. The lower electrode **12** may be formed by lift-off.
(34) As illustrated in FIG. **3**B and FIG. **3**E, the additional film **18***a* is formed on the first surface **15***a* of the piezoelectric film **14** so as to cover the tip portion of the lower electrode **12**. The additional film **18***a* is formed, for example, from the upper surface of the tip portion of the lower electrode **12** to the surface **15***a* in front of the lower electrode **12** of the piezoelectric film **14**. The additional film **18***a* is formed across the entire width of the lower electrode **12** and is formed to be longer than, for example, the width of the lower electrode **12**. The additional film **18***a* is formed by depositing a film by, for example, sputtering, vacuum evaporation, or CVD, and then patterning the film into a desired shape by photolithography and etching. The additional film **18***a* may be formed by lift-off.
(35) As illustrated in FIG. **3**C and FIG. **3**F, a sacrificial layer **70** covering the lower electrode **12** and the additional film **18***a* is formed on the surface **15***a* of the piezoelectric film **14**. The thickness of the sacrificial layer **70** is, for example, about 50 nm to 300 nm. The sacrificial layer **70** is selected from materials easily soluble in an etching solution or etching gas, such as MgO, ZnO, Ge, or SiO.sub.2. The sacrificial layer **70** is formed by depositing a film by, for example, sputtering, vacuum evaporation, or CVD, and then patterning the film into a desired shape by photolithography and etching. The sacrificial layer **70** may be formed by lift-off.
(36) As illustrated in FIG. **4**A and FIG. **4**D, the insulating film **26** covering the sacrificial layer **70** is formed on the first surface **15***a* of the piezoelectric film **14**, and the upper surface of the insulating film **26** is then polished by, for example, chemical mechanical polishing (CMP). As a result, the insulating film **26** has a desired thickness and the upper surface thereof is planarized.
(37) As illustrated in FIG. **4**B and FIG. **4**E, the insulating film **26** is bonded to the substrate **10**. That is, the multilayer structure is bonded to the substrate while the surface **15***a* side of the piezoelectric film **14** is opposite to the substrate **10**. Thereafter, the upper surface of the piezoelectric film **14** is polished by, for example, CMP to be thinned.
(38) As illustrated in FIG. **4**C and FIG. **4**F, the upper electrode **16** is formed on the other surface **15***b* (a second surface) of the piezoelectric film **14**. The upper electrode **16** is formed by depositing a film by, for example, sputtering, vacuum evaporation, or CVD, and then patterning the film into a desired shape by photolithography and etching. The upper electrode **16** may be formed by lift-off.
(39) As illustrated in FIG. **5**A and FIG. **5**C, the additional film **18***b* is formed on the surface **15***b* of the piezoelectric film **14** so as to cover the tip portion of the upper electrode **16**. The additional film **18***b* is formed so as to be away from the additional film **18***a* in the Y direction. The additional film **18***b* is formed, for example, from the upper surface of the tip portion of the upper electrode **16** to the surface **15***b* in front of the upper electrode **16** of the piezoelectric film **14**. The additional film **18***b* is formed across the entire width of the upper electrode **16** and is formed so as to be longer

than, for example, the width of the upper electrode **16**. The additional film **18***b* is formed by depositing a film by, for example, sputtering, vacuum evaporation, or CVD, and then patterning the film into a desired shape by photolithography and etching. The additional film **18***b* may be formed by lift-off.

(40) As illustrated in FIG. **5**B and FIG. **5**D, the passivation film **28** covering the upper electrode **16** and the additional film **18***b* is formed on the surface **15***b* of the piezoelectric film **14**. The passivation film **28** is formed by, for example, sputtering, vacuum evaporation, or CVD. Then, an aperture **72** is formed in the passivation film **28** on the upper electrode **16** by, for example, photolithography and etching. Then, the protective film **28** and the piezoelectric film **14** are machined by, for example, milling to form a pair of the through holes **22**, which sandwich the resonance region **50**, where the lower electrode **12** and the upper electrode **16** overlap, therebetween in the X direction and are located along the resonance region **50** in the Y direction, in the piezoelectric film **14**, and to form an aperture **74** dug up until reaching the lower electrode **12**. The through holes **22** may reach a part of the sacrificial layer **70**.

(41) As illustrated in FIG. **6**A and FIG. **6**C, the terminal electrode **24***a* electrically connected to the lower electrode **12** is formed in the aperture **74**, and the terminal electrode **24***b* electrically connected to the upper electrode **16** is formed in the aperture **72** by, for example, lift-off.

(42) As illustrated in FIG. **6**B and FIG. **6**D, an etching medium such as an etching solution or etching gas for the sacrificial layer **70** is introduced into the sacrificial layer **70** below the lower electrode **12** through the through holes **22**. This process removes the sacrificial layer **70**. As the etching medium for etching the sacrificial layer **70**, the medium that hardly etches materials constituting the resonator other than the sacrificial layer **70** is preferable. Particularly, the etching medium is preferably a medium that hardly etches the lower electrode **12** and the additional film **18***a*, which come in contact with the etching medium. By removing the sacrificial layer **70**, the air gap **30** is formed between the lower electrode **12** and the insulating film **26** and between the additional film **18***a* and the insulating film **26**. As described above, the piezoelectric thin film resonator **100** according to the first embodiment is formed.

(43) Here, the crystal orientations of the piezoelectric film **14** will be described. First, the definition of Euler angles (φ, θ, ψ) will be described. In a right-handed XYZ coordinate system, a normal direction of the upper surface of the piezoelectric layer **14** is defined as a Z direction, and directions orthogonal to the Z direction and orthogonal to each other in a plane direction of the upper surface of the piezoelectric layer **14** are defined as an X direction and a Y direction. First, the X direction, the Y direction, and the Z direction are configured to be correspond to the X-axis orientation, the Y-axis orientation, and the Z-axis orientation of the crystal orientations, respectively. Then, the +X direction is rotated by an angle φ about the Z direction from the +X direction to the +Y direction. The +Y direction is rotated by an angle θ about the X direction after the rotation of the angle φ from the +Y direction to the +Z direction. The +X direction is rotated by an angle ψ about the Z direction after the rotation of the angle θ from the +X direction to the +Y direction. The Euler angles obtained by rotating the directions in this manner are (φ, θ, ψ). The Euler angles expressed using (φ, θ, ψ) include equivalent Euler angles.

(44) FIG. **7**A and FIG. **7**B illustrate the crystal orientations of the piezoelectric film **14**. FIG. **7**A and FIG. **7**B illustrate a case in which the piezoelectric film **14** is a lithium niobate film or a lithium tantalate film. In FIG. **7**A and FIG. **7**B, broken line arrows on the left side indicate the orientations of the crystal axes of the piezoelectric film **14**. Solid line arrows on the right side correspond to the X direction, Y direction, and Z direction in FIG. **1**, FIG. **2**A, and FIG. **2**B. As illustrated in FIG. **7**A, the +X direction, the +Y direction, and +Z direction are made to correspond to the +X-axis orientation, +Y-axis orientation, and +Z-axis orientation of the crystal orientations of the piezoelectric film **14**, respectively. As illustrated in FIG. **7**B, from the state illustrated in FIG. **7**A, the Y direction and the Z direction are rotated 105° in the YZ plane about the X direction from the Y direction to the Z direction. In the case that the directions have been rotated as described above, a

direction obtained by rotating the +Z-axis orientation of the crystal orientations 105° toward the +Y-axis orientation corresponds to the +Z direction. In this case, the Y direction corresponds to the vibration direction **60** of the thickness-shear vibration. The Euler angles are (0°, −105°, 0°).

(45) The normal direction (the Z direction) of the upper surface of the piezoelectric film **14** is the direction in the Y-axis Z-axis plane. Therefore, thickness-shear vibration occurs in the planar direction of the piezoelectric film **14**. The X-axis orientation is preferably a direction at an angle within a range of ±5° from the planar direction of the piezoelectric film **14**, more preferably a direction at an angle within a range of ±1° from the planar direction of the piezoelectric film **14**. The normal direction (the Z direction) of the upper surface of the piezoelectric film **14** is set to a direction obtained by rotating the +Z-axis orientation of the crystal orientations by 105° toward the +Y-axis orientation. As a result, the vibration direction **60** of the thickness-shear vibration and the direction orthogonal thereto are the planar direction of the piezoelectric film **14**. The Z direction is preferably a direction at an angle within a range of ±5° from a direction obtained by rotating the +Z-axis orientation 105° from the +Z-axis orientation to the +Y-axis orientation, more preferably a direction at an angle within a range of ±1° from the direction obtained by rotating the Z-axis orientation 165° from the +Z-axis orientation to the +Y-axis orientation. The Euler angles are preferably (0°±5°, −105±5°, 0°±5°).

(46) FIG. **8** is a plan view of a piezoelectric thin film resonator **1000** in accordance with a comparative example. FIG. **9**A is a cross-sectional view taken along line A-A in FIG. **8**, and FIG. **9**B is a cross-sectional view taken along line B-B in FIG. **8**. In FIG. **8**, an additional film **118***b* is also hatched in addition to the resonance region **50** for clarity of the figure. As illustrated in FIG. **8**, FIG. **9**A, and FIG. **9**B, in the piezoelectric thin film resonator **1000** in accordance with the comparative example, a ring-shaped additional film **118***a* along the four sides of the resonance region **50** is provided under the piezoelectric film **14** and the lower electrode **12**, and a ring-shaped additional film **118***b* along the four sides of the resonance region **50** is provided on the piezoelectric film **14** and the upper electrode **16**. That is, the additional films **118***a* and **118***b* are provided in all of the following regions: the edge regions **54***a* and **54***b* located on respective sides of the resonance region **50** in the Y direction and edge regions **54***c* and **54***d* located on respective sides of the resonance region **50** in the X direction. The additional films **118***a* and **118***b* are provided from the edge regions **54***a* to **54***d* of the resonance region **50** to regions **56***a* to **56***d* outside the resonance region **50**. Through holes **122** for removing the sacrificial layer used to form the air gap **30** is provided outside the additional films **118***a* and **118***b* so as to be away from the resonance region **50**. Other configurations are the same as those of the first embodiment, and the description thereof is thus omitted.

(47) In the comparative example, the ring-shaped additional films **118***a* and **118***b* are provided in the edge regions **54***a* to **54***d* of the resonance region **50**. Therefore, the acoustic velocity of the acoustic wave in the edge regions **54***a* to **54***d* becomes lower than the acoustic velocity of the acoustic wave in the central region **52**, and the piston mode is achieved. Thus, lateral-mode spurious emissions due to the acoustic waves propagating in the X direction and the Y direction are reduced. However, since the through holes **122** for removing the sacrificial layer are provided further outward than the additional films **118***a* and **118***b*, the piezoelectric thin film resonator **1000** becomes large in size.

(48) When an etching medium is introduced from one of a pair of the through holes **122** to the other to remove the sacrificial layer, as illustrated in FIG. **9**A, the additional film **118***a* is present in the traveling direction of the etching medium. Therefore, it is difficult to remove the sacrificial layer formed in the inner region surrounded by the additional film **118***a*, and the sacrificial layer may remain. When the sacrificial layer remains and, for example, a hollow structure is not be obtained, the characteristics are deteriorated.

(49) In contrast, in the first embodiment, as illustrated in FIG. **1**, FIG. **2**A, and FIG. **2**B, the additional films **18***a* and **18***b* are provided in the edge regions **54***a* and **54***b* located on respective

sides of the central region **52** of the resonance region **50** in the Y direction (a second direction). The through holes **22** are connected to the air gap **30**, sandwich the resonance region **50** therebetween in the X direction (a first direction), and is provided along the resonance region **50**. Therefore, the piezoelectric thin film resonator **100** can be miniaturized. Moreover, lateral-mode spurious emissions due to the acoustic wave propagating in the Y direction are reduced by the additional films **18***a* and **18***b*. Lateral-mode spurious emissions due to the acoustic wave propagating in the X direction are reduced by forming free ends by the through holes **22** provided along the resonance region **50**. Therefore, in the first embodiment, it is possible to reduce the size of the piezoelectric thin film resonator **100** while reducing lateral-mode spurious emissions. As described above, the through holes **22** have two roles: removing the sacrificial layer used for forming the air gap **30** and reducing lateral-mode spurious emissions. The additional films **18***a* and **18***b* may be provided in edge regions located on both sides of the central region **52** of the resonance region **50** in a direction substantially orthogonal to the direction in which the through holes **22** sandwich the resonance region **50** therebetween. The term "substantially orthogonal" means that the inclination from the orthogonal direction to the extent of manufacturing errors is acceptable, and the inclination with respect to the orthogonal direction within a range of ±5° is acceptable.

(50) In addition, in the first embodiment, as illustrated in FIG. **3**B and FIG. **3**E, the additional film **18***a* (a first additional film) having a substantially rectangular shape in a plan view is formed on at least the tip portion of the lower electrode **12**. As illustrated in FIG. **3**C and FIG. **3**F, the sacrificial layer **70** covering the lower electrode **12** and the additional film **18***a* is formed on the surface **15***a* of the piezoelectric film **14**. As illustrated in FIG. **4**B and FIG. **4**E, the multilayer structure is bonded to the substrate **10** while the surface **15***a* side of the piezoelectric film **14** is opposite to the substrate **10**. As illustrated in FIG. **5**A and FIG. **5**C, the additional film **18***b* (a second additional film) having a substantially rectangular shape in a plan view is formed on at least the tip portion of the upper electrode **16**. As illustrated in FIG. **5**B and FIG. **5**D, a pair of the through holes **22**, which sandwich the resonance region **50** therebetween in the X direction (a first direction) and are provided along the resonance region **50**, is formed in the piezoelectric film **14**. As illustrated in FIG. **6**B and FIG. **6**D, the sacrificial layer **70** is removed by introducing an etching medium into the pair of the through holes **22**. As illustrated in FIG. **6**B, since the additional film **18***a* does not exist in the traveling direction of the etching medium introduced into the pair of the through holes **22**, the sacrificial layer **70** can be prevented from remaining without being removed. The substantially rectangular shape of the additional films **18***a* and **18***b* in a plan view is not limited to a case in which each side is completely straight, and a case in which the shape is deformed from a rectangle to the extent that does not affect the device characteristics is also acceptable.

(51) FIG. **10**A to FIG. **10**C are plan views illustrating other examples of the air gap **30**. FIG. **1** illustrates a case in which the air gap **30** is substantially rectangular in a plan view as an example, but this does not intend to suggest any limitation. The air gap **30** may have a circular shape as illustrated in FIG. **10**A, an elliptical shape as illustrated in FIG. **10**B, or a polygonal shape having five or greater apexes such as a hexagonal shape as illustrated in FIG. **10**C. The substantially rectangular shape of the gap **30** in a plan view is not limited to a case in which each side is completely straight, and a case in which the shape is deformed from a rectangle to the extent that does not affect the device characteristics is also acceptable.

(52) By forming the lower electrode **12** on the surface **15***a* of the piezoelectric film **14** by sputtering or the like and forming the upper electrode **16** on the surface **15***b* of the piezoelectric film **14** by sputtering or the like, stresses that the piezoelectric film **14** receives from the lower electrode **12** and the upper electrode **16** are cancelled. Therefore, when the sacrificial layer **70** is removed, the lower electrode **12** and/or the additional film **18***a* may stick to the sacrificial layer **70**. This sticking tends to occur at the corners of the air gap **30**. Therefore, by forming the air gap **30** in a circular shape or an elliptical shape, or by forming the air gap **30** in a polygonal shape having large apex angles, it is possible to inhibit the sticking between the lower electrode **12** and/or the additional

film **18***a* and the sacrificial layer **70** at the corner portion of the air gap **30**.

(53) In the first embodiment, the X direction in which a pair of the through holes **22** sandwich the resonance region **50** therebetween is a direction substantially orthogonal to the vibration direction **60** of the thickness-shear vibration. The additional films **18***a* and **18***b* are provided in the edge regions **54***a* and **54***b* of the resonance region **50** in the Y direction that is substantially parallel to the vibration direction **60** of the thickness-shear vibration, respectively. As described above, the through holes **22** are provided so as to sandwich the resonance region **50** therebetween in the direction substantially orthogonal to the vibration direction **60** of the thickness-shear vibration, and thus it is possible to effectively reduce lateral-mode spurious emissions due to the acoustic wave propagating in the direction substantially orthogonal to the vibration direction **60** of the thickness-shear vibration. The direction substantially orthogonal to the vibration direction **60** of the thickness-shear vibration means that a case in which the inclination from the direction substantially orthogonal to the vibration direction **60** of the thickness-shear vibration to the extent of manufacturing errors is acceptable, and a case in which the inclination from the direction substantially orthogonal to the vibration direction **60** of the thickness-shear vibration within a range of ±5° is acceptable. Further, by providing the additional films **18***a* and **18***b* in the edge regions **54***a* and **54***b* in the direction that is substantially parallel to the vibration direction **60** of the thickness-shear vibration, it is possible to reduce lateral-mode spurious emissions due to the acoustic wave propagating in the direction substantially parallel to the vibration direction **60** of the thickness-shear vibration. The direction substantially parallel to the vibration direction **60** of the thickness-shear vibration means that a case in which the inclination from the vibration direction **60** of the thickness-shear vibration to the extent of manufacturing errors is acceptable, and a case in which the inclination from the vibration direction **60** of the thickness-shear vibration within a range of ±5° is acceptable.

(54) In the first embodiment, as illustrated in FIG. **1**, the substantially entire circumference of the resonance region **50** is surrounded by the additional films **18***a* and **18***b* and a pair of the through holes **22** in a plan view. This structure effectively reduces lateral-mode spurious emissions. The substantially entire circumference is 90% or greater of the circumference of the resonance region **50**, and may be 95% or greater of the circumference of the resonance region **50**. In a plan view, ½ or greater of, or ¾ or greater of the circumference of the resonance region **50** may be surrounded by the additional films **18***a* and **18***b* and a pair of the through holes **22**. The entire circumference of the resonance region **50** is preferably completely surrounded by the additional films **18***a* and **18***b* and a pair of the through holes **22** in a plan view.

(55) In the first embodiment, as illustrated in FIG. **1**, in a plan view, a pair of the through holes **22** are in contact with the resonance region **50** on respective sides of the resonance region **50** in the X direction (a first direction). Thus, lateral-mode spurious emissions are effectively reduced because of the formation of the free ends by the through holes **22**.

(56) In the first embodiment, as illustrated in FIG. **1**, the resonance region **50** has a substantially rectangular shape in a plan view, and the sides of the resonance region **50** opposite to each other in the X direction (a first direction) are defined by a pair of the through holes **22**. Thereby, it is possible to effectively reduce lateral-mode spurious emissions due to the acoustic wave propagating in the X direction. The term “substantially rectangular” is not limited to a case in which each side is completely straight, and a case in which the shape is deformed from a rectangle to the extent that does not affect the device characteristics is also acceptable.

(57) Furthermore, in the first embodiment, as illustrated in FIG. **1**, the additional films **18***a* and **18***b* are longer than the resonance region **50** in the X direction (a first direction), and at least a part of each of the through holes **22** is sandwiched between the additional films **18***a* and **18***b* in the Y direction (a second direction) in a plan view. Thus, even when the positions of the additional films **18***a* and **18***b* are shifted because of manufacturing errors, it is possible to obtain a structure in which the substantially entire circumference of the resonance region **50** is surrounded by the

additional films **18***a* and **18***b* and a pair of the through holes **22** in a plan view. Therefore, lateral-mode spurious emissions can be effectively reduced. The additional films **18***a* and **18***b* may be provided to have substantially the same length as the resonance region **50** in the X direction. The term "substantially the same length" means that a difference is acceptable to the extent of manufacturing errors, and the ratio of one length to the other length may be 95% or more and 105% or less.

Second Embodiment

(58) FIG. **11** is a plan view of a piezoelectric thin film resonator **200** in accordance with a second embodiment. FIG. **12**A is a cross-sectional view taken along line A-A in FIG. **11**, and FIG. **12**B is a cross-sectional view taken along line B-B in FIG. **11**. As illustrated in FIG. **11**, FIG. **12**A, and FIG. **12**B, in the piezoelectric thin film resonator **200** in accordance with the second embodiment, the additional film **18***a* is provided in each of the edge regions **54***a* and **54***b*, which are located on respective sides of the resonance region **50** in the Y direction, in a substantially rectangular shape in a plan view. Similarly, the additional film **18***b* is provided in each of the edge regions **54***a* and **54***b*, which are located on respective sides of the resonance region **50** in the Y direction, in a substantially rectangular shape in a plan view. Other configurations are the same as those in the first embodiment, and the description thereof is thus omitted.

(59) The additional film **18***a* may be provided only in the edge region **54***a* as in the first embodiment, or may be provided in the edge region **54***a* and the edge region **54***b* as in the second embodiment. Similarly, the additional film **18***b* may be provided only in the edge region **54***b* as in the first embodiment, or may be provided in the edge region **54***a* and the edge region **54***b* as in the second embodiment.

(60) The second embodiment is not limited to the case in which both of the additional films **18***a* and **18***b* are provided, and may be applied to the case in which only one of the additional films **18***a* and **18***b* is provided. The positions of the additional films **18***a* and **18***b* in the Y direction in the edge regions **54***a* and **54***b* are required to have high precision to reduce lateral-mode spurious emissions. To achieve high positional precision, as in the first embodiment, it is preferable that the additional film **18***a* be provided in the edge region **54***a*, which is the tip portion of the lower electrode **12**, and the additional film **18***b* be provided in the edge region **54***b*, which is the tip portion of the upper electrode **16**.

Third Embodiment

(61) FIG. **13** is a plan view of a piezoelectric thin film resonator **300** in accordance with a third embodiment. As illustrated in FIG. **13**, in the piezoelectric thin film resonator **300** in accordance with the third embodiment, the additional film **18***a* extends from the resonance region **50** in the +X direction, and does not extend from the resonance region **50** in the −X direction. Unlike the additional film **18***b*, the additional film **18***a* extends from the resonance region **50** in the −X direction and does not extend from the resonance region **50** in the +X direction. Of a pair of the through holes **22**, the through hole **22** located at the +X direction side extends from the resonance region **50** in the −Y direction, and the through hole **22** located at the −X direction side extends from the resonance region **50** in the +Y direction. Other configurations are the same as those in the first embodiment, and the description thereof is thus omitted.

(62) In the third embodiment, the length of each of a pair of the through holes **22** in the Y direction (a second direction) is longer than the length of the resonance region **50** in the Y direction (the second direction). Thus, when an etching medium is introduced into the through holes **22** to remove the sacrificial layer **70**, the sacrificial layer **70** can be efficiently removed.

(63) In the third embodiment, the lengths in the Y direction of both of the pair of the through holes **22** are longer than the length in the Y direction of the resonance region **50**. However, the length in the Y direction of at least one of the pair of the through holes **22** may be longer than the length in the Y direction of the resonance region **50**. In addition, both of the pair of the through holes **22** may extend from the resonance region **50** in the +Y direction or the −Y direction.

Fourth Embodiment

(64) FIG. **14**A and FIG. **14**B are cross-sectional views of a piezoelectric thin film resonator **400** in accordance with a fourth embodiment. As illustrated in FIG. **14**A and FIG. **14**B, in the piezoelectric thin film resonator **400** in accordance with the fourth embodiment, the side surface of the additional film **18***a* is a tapered surface that is inclined so that the additional film **18***a* becomes wider toward the piezoelectric film **14** in a cross-sectional view. An angle θ of the side surface of the additional film **18***a* with respect to a normal line **90** of the surface of the piezoelectric film **14** on which the lower electrode **12** is provided is, for example, about 5° to 40°. Other configurations are the same as those in the first embodiment, and the description thereof is thus omitted.

(65) In the fourth embodiment, the side surface of the additional film **18***a* is a tapered surface that is inclined with respect to the normal line **90** of the surface of the piezoelectric film **14** on which the lower electrode **12** is provided, in a cross-sectional view. It is desirable to thin the sacrificial layer **70** to inhibit the breakage of the piezoelectric film **14**. However, when the sacrificial layer **70** is thinned, the sacrificial layer **70** may rapture on the side surface of the additional film **18***a*. By forming the side surface of the additional film **18***a* as a tapered surface, it is possible to inhibit the rupture of the sacrificial layer **70**.

Fifth Embodiment

(66) FIG. **15** is a plan view of a piezoelectric thin film resonator **500** in accordance with a fifth embodiment. FIG. **16**A is a cross-sectional view taken along line A-A in FIG. **15**, and FIG. **16**B is a cross-sectional view taken along line B-B in FIG. **15**. In FIG. **15**, for clarity of the drawing, the additional film **18***b* is also hatched in addition to the resonance region **50**. As illustrated in FIG. **15**, FIG. **16**A, and FIG. **16**B, in the piezoelectric thin film resonator **500** in accordance with the fifth embodiment, the additional film **18***a* is provided in a ring shape under the piezoelectric film **14**, and the additional film **18***b* is provided in a ring shape on the piezoelectric film **14**. The through holes **22** are provided inside the ring-shaped additional films **18***a* and **18***b*. Other configurations are the same as those in the first embodiment, and the description thereof is thus omitted.

(67) In the fifth embodiment, the additional films **18***a* and **18***b* are provided in a ring shape, and the through holes **22** are provided inside the ring-shaped additional films **18***a* and **18***b*. Accordingly, it is possible to obtain a structure in which the entire circumference of the resonance region **50** is surrounded by the additional films **18***a* and **18***b* and a pair of the through holes **22** in a plan view. Therefore, lateral-mode spurious emissions can be effectively reduced.

(68) The fifth embodiment has described the case in which the additional film **18***a* is provided under the lower electrodes **12** and the additional film **18***b* is provided on the upper electrode **16** as an example, but only one of the additional films **18***a* and **18***b* may be provided.

(69) The first to fifth embodiments have described the case in which the thickness-shear vibration is excited in the resonance region **50** as an example, but the thickness extension vibration may be excited. In addition, the shape of the resonance region **50** is not limited to a substantially rectangular shape in a plan view, and may be another shape such as a polygonal shape having five or greater apexes or an elliptical shape.

Sixth Embodiment

(70) FIG. **17** is a circuit diagram of a filter **600** in accordance with a sixth embodiment. As illustrated in FIG. **17**, in the filter **600**, one or more series resonators S**1** to S**4** are connected in series between an input terminal T**1** and an output terminal T**2**. One or more parallel resonators P**1** to P**3** are connected in parallel between the input terminal T**1** and the output terminal T**2**. The piezoelectric thin film resonator according to any one of the first to fifth embodiments can be used as at least one of the following resonators: one or more series resonators S**1** to S**4** and one or more parallel resonators P**1** to P**3**. The number of resonators in the ladder-type filter can be set as appropriate.

Seventh Embodiment

(71) FIG. **18** is a block diagram of a duplexer **700** in accordance with a seventh embodiment. As

illustrated in FIG. **18**, in the duplexer **700**, a transmit filter **40** is connected between a common terminal Ant and a transmit terminal Tx. A receive filter **42** is connected between the common terminal Ant and a receive terminal Rx. The transmit filter **40** transmits signals in the transmit band to the common terminal Ant as transmission signals among signals input from the transmit terminal Tx, and suppresses signals with other frequencies. The receive filter **42** transmits signals in the receive band to the receive terminal Rx as reception signals among signals input from the common terminal Ant, and suppresses signals with other frequencies. Either or both of the transmit filter **40** and the receive filter **42** may be the filter **600** of the sixth embodiment.
(72) In the seventh embodiment, a duplexer is described as an example of a multiplexer, but the multiplexer may be a triplexer or a quadplexer.
(73) Although the embodiments of the present invention have been described in detail, it is to be understood that the various change, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. An acoustic wave device comprising: a substrate; a lower electrode and an upper electrode provided over the substrate; a piezoelectric film that is provided over the substrate, at least a part of the piezoelectric film being interposed between the lower electrode and the upper electrode thereby defining a resonance region in a plan view where the lower electrode and the upper electrode overlap with each other across said at least a part of the piezoelectric film, the piezoelectric film having a pair of through holes, the pair of through holes sandwiching the resonance region therebetween in a first direction in the plan view, being provided along the resonance region, and being connected to an air gap, the air gap being formed between the substrate and the lower electrode and overlapping with the resonance region in the plan view; and additional films that are not provided in a central region of the resonance region in the plan view and that are not provided in entire respective first edge regions, which are located on respective sides of the central region in the first direction in the plan view, of the resonance region, the additional film being provided in respective second edge regions, which are located on respective sides of the central region in a second direction substantially orthogonal to the first direction in the plan view, of the resonance region.
2. The acoustic wave device according to claim 1, wherein the lower electrode and the upper electrode excite thickness-shear vibration in the piezoelectric film, and wherein the second direction is substantially parallel to a vibration direction of the thickness-shear vibration.
3. The acoustic wave device according to claim 1, wherein in the plan view, the resonance region is surrounded by the pair of through holes in the first direction and by the additional films in the second direction, so that a substantially entire circumference of the resonance region is surrounded by the additional films and the pair of through holes.
4. The acoustic wave device according to claim 1, wherein, in the plan view, the pair of through holes are in contact with the resonance region at respective sides of the resonance region in the first direction.
5. The acoustic wave device according to claim 1, wherein the resonance region has a substantially rectangular shape in the plan view, and sides opposite to each other in the first direction of the resonance region are defined by the pair of through holes.
6. A filter comprising: the acoustic wave device according to claim 1.
7. A multiplexer comprising: the filter according to claim 6.
8. An acoustic wave device comprising: a substrate; a lower electrode and an upper electrode provided over the substrate; a piezoelectric film that is provided over the substrate, at least a part of the piezoelectric film being interposed between the lower electrode and the upper electrode thereby defining a resonance region in a plan view where the lower electrode and the upper electrode

overlap with each other across said at least a part of the piezoelectric film, the piezoelectric film having a pair of through holes, the pair of through holes sandwiching the resonance region therebetween in a first direction in the plan view, being provided along the resonance region, and being connected to an air gap, the air gap being formed between the substrate and the lower electrode and overlapping with the resonance region in the plan view; and additional films that are not provided in a central region of the resonance region in the plan view and are provided in respective edge regions, which are located on respective sides of the central region in a second direction substantially orthogonal to the first direction in the plan view, of the resonance region, wherein the additional films are longer than the resonance region in the first direction, and wherein at least a part of each of the pair of through holes is sandwiched between the additional films in the second direction in the plan view.

9. An acoustic wave device comprising: a substrate; a lower electrode and an upper electrode provided over the substrate; a piezoelectric film that is provided over the substrate, at least a part of the piezoelectric film being interposed between the lower electrode and the upper electrode thereby defining a resonance region in a plan view where the lower electrode and the upper electrode overlap with each other across said at least a part of the piezoelectric film, the piezoelectric film having a pair of through holes, the pair of through holes sandwiching the resonance region therebetween in a first direction in the plan view, being provided along the resonance region, and being connected to an air gap, the air gap being formed between the substrate and the lower electrode and overlapping with the resonance region in the plan view; and additional films that are not provided in a central region of the resonance region in the plan view and are provided in respective edge regions, which are located on respective sides of the central region in a second direction substantially orthogonal to the first direction in the plan view, of the resonance region, wherein the additional films are provided in a ring shape on at least one of the following surfaces: a first surface of the piezoelectric film on which the lower electrode is provided and a second surface of the piezoelectric film on which the upper electrode is provided, and wherein the pair of through holes are provided inside the ring-shaped additional films.